Geospatial Explainable AI for Residential Property Investment Intelligence in Hyderabad and Telangana: System Architecture, Econometric Grounding, and Academic Defense Blueprint

Executive System Paradigm and Architectural Blueprint

Urban real estate markets in rapidly industrializing metropolitan corridors suffer from severe information asymmetry, fragmented pricing disclosures, and opaque valuation heuristics[1]. While consumer navigation platforms have standardized spatial decision-making by rendering complex vehicular telemetry into intuitive green-to-red choropleth overlays, residential property analysis in Indian metropolitan regions remains trapped in static, listing-by-listing search paradigms[2]. The proposed system resolves this analytical gap by introducing an interactive geospatial investment intelligence platform for Hyderabad and Telangana that functions analogously to traffic-color mapping for capital allocation[5]. Rather than operating as a transactional brokerage portal, the platform functions strictly as an automated valuation and decision-support engine, shading each urban and peri-urban locality along a continuous green-to-red spectrum corresponding to a composite 1-to-10 investment score[4].

This deliberate boundary condition—excluding property listing brokerage, buyer-seller negotiation workflows, and payment processing—serves as an essential architectural safeguard. By isolating predictive analytics, spatial econometrics, and explainable artificial intelligence (XAI) from commercial transaction execution, the engineering footprint remains mathematically rigorous, computationally tractable, and insulated from the regulatory liabilities associated with real estate intermediation[1].

The Five-Screen User Experience Architecture

The front-end presentation layer translates high-dimensional spatial and tabular modeling into a progressive disclosure interface structured across five distinct analytical views[4]. Each view corresponds to a specific stage in an investor's cognitive workflow, transitioning from macro-spatial screening to micro-level attribution and cross-border net return simulation[7].

Screen Identifier	Interface	Core Visual &	Primary

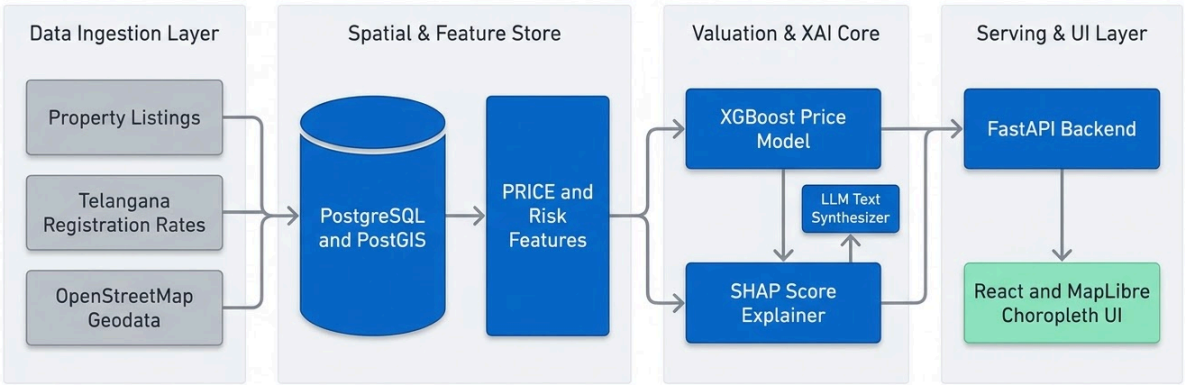
	Component	Analytical Elements	Decision-Support Function
Screen 1	**Main Choropleth Map**	Full-viewport MapLibre GL canvas shading Hyderabad and Telangana localities from green (high investment score) to red (low score); interactive filter chips for **Budget**, **Investment Goal** (*Rent*, *Resale*, *Long-Term*), and **Property Type**.	Enables rapid macro-spatial screening and dynamic re-weighting of locality scores based on capital constraints and horizon preferences[5].
Screen 2	**Locality Intelligence Panel**	Slide-over drawer triggered upon polygon selection displaying the composite score (scaled 1 to 10), estimated price per square foot, a 3-year historical price trajectory line, and summary contribution bars.	Provides immediate quantitative context on baseline valuation levels, historical momentum, and primary score drivers for a selected micro-market[4].
Screen 3	**"Why This Score" XAI Attribution View**	Diverging horizontal bar chart mapping local Shapley Additive Explanations (SHAP) into signed score adjustments (e.g., $+2.1$ *near upcoming metro*,	Eliminates the "black-box" trust deficit of machine learning valuations by exposing the exact additive positive and negative drivers behind a locality's rating[1].

		-1.3 *high baseline price*), paired with a single plain-English explanatory sentence generated via an LLM translation layer.	
Screen 4	**NRI Net Return Calculator**	Parametric financial simulator accepting **Budget**, **Foreign Currency**, and **Holding Period (Years)**; outputs net annualized return after accounting for rental cash flows, capital appreciation, Indian taxation, and foreign exchange depreciation.	Quantifies real repatriable returns for diaspora investors by modeling statutory transaction costs, post-Budget 2024 capital gains taxes, and currency erosion[9].
Screen 5	**Multi-Locality Comparison Matrix**	Side-by-side tabular comparison of 2 to 3 selected localities across predicted price, 3-year growth, rental yield, infrastructure proximity, employment accessibility, risk penalties, and net NRI yield.	Facilitates final relative trade-off analysis between competing micro-markets across the metropolitan gradient[4].

End-to-End Spatial and Tabular Processing Pipeline

Beneath the five-screen user interface, the internal data and modeling architecture executes a six-stage sequential pipeline: raw multi-source ingestion, spatial cleaning and polygon binding,

domain-specific PRICE feature engineering, gradient-boosted regression, game-theoretic explainability decomposition, and vector map rendering[4].



Each component within this internal technology stack is selected to optimize tabular predictive accuracy, spatial query throughput, and algorithmic interpretability[5]. Because real estate valuation is intrinsically spatial, standard relational databases cannot efficiently compute geodesic distances, buffer intersections, or point-in-polygon joins[7]. Integrating PostgreSQL with the PostGIS spatial extension enables native execution of geometric operations—such as calculating the exact Euclidean or road-network distance from a locality centroid to the nearest Hyderabad Metro station, Outer Ring Road (ORR) interchange, or employment polygon—directly within the database layer[6]. Once spatial features are attached to cleaned property records, the predictive valuation engine relies on XGBoost or LightGBM[13]. Residential property datasets are predominantly tabular, characterized by non-linear feature interactions, skewed price distributions, and heterogeneous measurement scales[5]. Gradient-boosted decision trees consistently outperform both parametric linear regressions and deep neural networks on structured real estate datasets of moderate sample size while maintaining fast training cycles[13].

To resolve the opacity of tree ensembles, the pipeline integrates SHAP (Shapley Additive Explanations), which decomposes every locality's predicted valuation and composite score into exact additive marginal contributions from each PRICE input variable[4]. These Shapley values directly drive the diverging horizontal bar chart in the explanation panel[4]. Because raw numerical attributions can remain technical for retail users, an optional Large Language Model (LLM) wording layer translates the highest-magnitude SHAP feature-value pairs into a single, deterministic plain-English sentence without permitting the language model to alter or hallucinate the underlying calculations[4]. Finally, an asynchronous FastAPI backend serves

pre-computed GeoJSON feature collections and executes on-the-fly NRI tax and currency simulations, while a React and MapLibre GL frontend utilizes hardware-accelerated vector tiles to render smooth choropleth regions that re-color dynamically as filter chips are toggled[20].

System Part	Technology Choice	Architectural & Algorithmic Justification
Price Model	XGBoost or LightGBM	Superior predictive accuracy ($R^2 \approx$) on non-linear tabular housing data with rapid training convergence and native missing-value handling[4].
Explanation Engine	SHAP (TreeSHAP)	Computes exact game-theoretic additive feature attributions that map directly to the positive and negative "Why this score" adjustment bars[4].
Locality Score	Model Output + PRICE Factors ($1–10$)	Combines hedonic valuation residuals with forward-looking growth, yield, infrastructure, and risk vectors on an intuitive 1-to-10 scale[5].
Backend & GIS Store	FastAPI + PostgreSQL / PostGIS	High-concurrency asynchronous Python API paired with native spatial indexing (R-tree over GiST) for polygon and distance queries[16].
Frontend & Mapping	React + MapLibre GL	Open-source WebGL vector map rendering smooth green-to-red locality polygons and interactive filter states

		without proprietary tile paywalls.
Natural Language Polish	Constrained LLM Layer	Converts structured SHAP numerical tuples into a single human-readable summary sentence for immediate comprehension[4].

Econometric and Spatial Feature Engineering: The PRICE + Risk Framework

In hedonic pricing theory, first formalized by Rosen (1974), a differentiated asset is valued as an aggregation of the implicit shadow prices of its utility-bearing characteristics[5]. Contemporary explainable valuation frameworks express observed market value Y as an additive decomposition of economic (X_{ECO}), structural (X_{STR}), accessibility (X_{ACC}), and market-related (X_{MKT}) drivers[5]:

$$Y = f_{ECO}(X_{ECO}) + f_{STR}(X_{STR}) + f_{ACC}(X_{ACC}) + f_{MKT}(X_{MKT}) + \varepsilon$$

To operationalize this theoretical foundation into a domain-specific scoring rubric for Hyderabad and Telangana, the platform structures its spatial-economic variables around the **PRICE** framework, augmented by an explicit **Risk Penalty** subtractor[5].

Mathematical Formulation of the PRICE + Risk Index

For any given locality $l \in \{1, \ldots, N\}$, investment attractiveness is governed by five positive structural pillars and one negative risk adjustment[5]:

Mnemonic	Factor Dimension	Core Analytical Question	Engineered Spatial & Econometric Features
P	**Price Growth**	*Has the area been rising, and how fast?*	3-year Compound Annual Growth Rate ($CAGR_{3Y}$) of median price per

			sq.ft; ratio of retail asking price to Telangana government guidance unit rate[6]; price momentum relative to the GHMC metropolitan mean.
R	**Rental Yield**	*How much rent is earned per rupee invested?*	Annualized median monthly rent divided by median capital value ($\text{Yield}_l =$); tenant absorption velocity; compact 1BHK/2BHK vs. 3BHK rental premium ratio[19].
I	**Infrastructure**	*Are metro, roads, or ORR access coming?*	Binary and distance-decayed proximity indicators for Hyderabad Metro Phase-II corridors, Regional Ring Road (RRR) alignments, newly commissioned flyovers, and Strategic Road Development Plan (SRDP) nodes[14].
C	**Connectivity**	*How far from the metro, highways, and airport?*	Geodesic and network travel distance (km) and travel time via OpenStreetMap to the nearest operational Metro

			station, Outer Ring Road (ORR) exit interchange, railway terminal, and Shamshabad airport[6].
E	**Employment Hubs**	*How close are HITEC City, Financial District, and other job centers?*	Gravity-weighted accessibility index to primary employment centroids: HITEC City / Madhapur, Financial District / Gachibowli / Kokapet, Genome Valley (north), and Pocharam / Uppal IT parks (east)[6].
– Risk	**Risk Penalty**	*Do flood hazards, legal disputes, or oversupply impair capital safety?*	Deductions for low-lying flood-prone topography or lake buffer encroachments, active legal or land-title dispute density, and unsold inventory overhang (oversupply ratio)[9].

When a user toggles a specific **Goal** filter chip on Screen 1 (*Rent*, *Resale*, or *Long-Term*), the backend dynamically adjusts the convex combination weights $(\alpha_P, \alpha_R, \alpha_I, \alpha_C, \alpha_E, \gamma_{\text{Risk}})$ applied to the standardized factor sub-scores $(Z_P, Z_R, Z_I, Z_C, Z_E, Z_{\text{Risk}})$ before mapping the composite output onto the $[1, 10]$ interval:

$$\text{Score}_l^{(g)} = \text{clip}_{[1,10]}\left(1 + 9 \times \frac{S_l^{(g)} - S_{\min}^{(g)}}{S_{\max}^{(g)} - S_{\min}^{(g)}}\right)$$

Here, the unscaled goal-specific utility $S_l^{(g)}$ combines the machine learning model's hedonic valuation residual $\hat{\Delta}_{\text{val},l}$ (which measures whether a locality's current market price is undervalued relative to its structural and spatial fundamentals) with the weighted PRICE pillars minus the risk deduction[5]:

$$S_l^{(g)} = w_0 \hat{\Delta}_{\text{val},l} + \alpha_P^{(g)} Z_P + \alpha_R^{(g)} Z_R + \alpha_I^{(g)} Z_I + \alpha_C^{(g)} Z_C + \alpha_E^{(g)} Z_E - \gamma_{\text{Risk}} Z_{\text{Risk}}$$

Under a *Rental* goal, α_R and α_E receive dominant weights, prioritizing immediate cash yield and proximity to white-collar tenant pools in HITEC City and the Financial District[6]. Conversely, under a *Long-Term* horizon, α_I and α_P dominate, rewarding peri-urban corridors near upcoming metro extensions or ORR interchanges before asking prices fully capitalize the new infrastructure[6].

Spatial Reality of the Telangana Market and Dual-Source Price Anchoring

A recurring methodological flaw in academic house-price studies in India is an exclusive reliance on scraped commercial property portals[4]. Online portals record seller *asking prices* rather than *transacted sale prices*, introducing upward negotiation bias, duplicate broker listings, and inconsistent attribute reporting[2]. Furthermore, automated scraping of commercial portals frequently violates platform terms of service.

To construct a defensible dataset, the architecture anchors its curated residential market records against official **Telangana Registration and Stamps Department unit rates (Market Value Guidelines)** published on registration.telangana.gov.in[6]. Under Section 47-A of the Indian Stamp Act and the Telangana Market Value Guidelines Rules, 1998, statutory committees establish minimum unit rates per square yard for open plots and per square foot for built-up structures across every district, mandal, village, and urban locality in the state[6]. Because stamp duty and registration charges are legally levied on the higher of the deed consideration or the government market value, the statutory unit rate establishes an official valuation floor across Telangana[6].

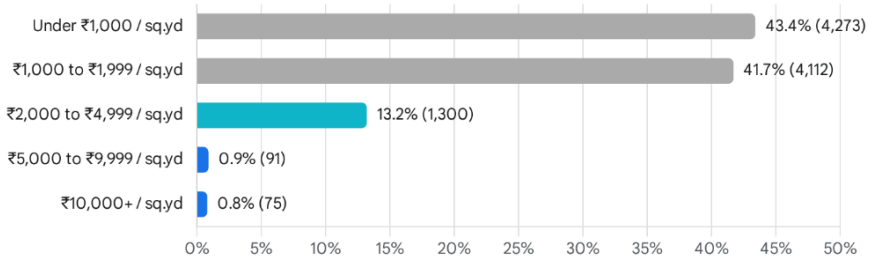
MACRO-TO-MICRO SPATIAL VALUATION ANALYSIS

Telangana Valuation Premium Is Concentrated in **Hyderabad and the Western IT Arc**

Panel 1: Statewide Distribution of Villages by Official Rate

Share (%) of 9,851 registered villages

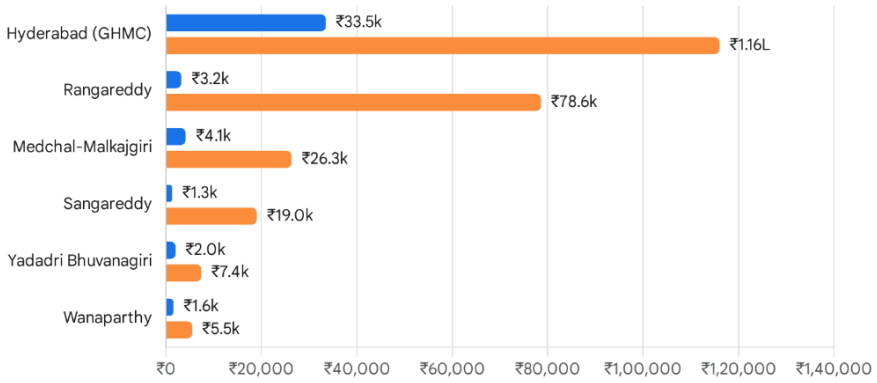
85.1% of Telangana villages sit under ₹2,000/sq.yd. Only 1.7% (~166 villages) reach ₹5,000/sq.yd or higher.



Panel 2: District-Level Valuation Cliff

Median Rate (₹/sq.yd) Peak Village Rate (₹/sq.yd)

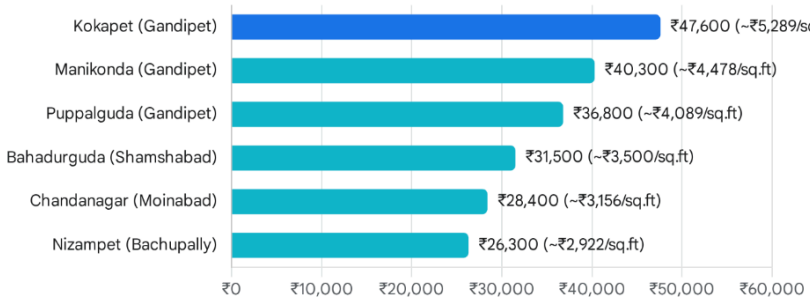
GHMC median rate is ~30x the state median (₹1,100). The capital corridor exhibits massive intra-district valuation spreads.



Panel 3: Western IT Corridor Micro-Market Leaders

Corridor Peak Top Suburban Arc

Median guidance rate per square yard across key localities in Gandipet, Shamshabad, Moinabad, and Bachupally.



Official Telangana guidance values reveal a steep urban-peri-urban cliff: while **85.1% of villages** state-wide are rated below Rs 2,000 per square yard, GHMC commands a median of Rs 33,500 per square yard, with western IT corridor hubs like Kokapet reaching Rs 47,600 per square yard.

Data sources: Telangana Registration Dept via Assetly Guidance Value Explorer

An empirical examination of the 2026 Telangana guidance value dataset reveals a sharp spatial concentration that directly justifies the sampling strategy for the Minimum Viable Product (MVP)[6]. Across the state of Telangana, the median open-plot guidance rate is approximately ₹1,100 per square yard, with 43.4% of villages (4,273 villages) rated below ₹1,000 per square yard and another 41.7% (4,112 villages) rated between ₹1,000 and ₹1,999 per square yard[6]. Only 1.7% of villages statewide—roughly 166 villages—exceed ₹5,000 per square yard[6]. By contrast, within the Greater Hyderabad Municipal Corporation (GHMC) core, the median official rate jumps thirty-fold to ₹33,500 per square yard, peaking at ₹1,16,000 per square yard in prime central localities[6].

Immediately beyond the GHMC boundary, virtually the entire metropolitan valuation premium sits within three enveloping districts: **Medchal-Malkajgiri** (district median ₹4,100/sq.yd, peaking at ₹26,300/sq.yd), **Rangareddy** (district median ₹3,200/sq.yd, peaking at ₹78,600/sq.yd in its western IT mandals), and **Sangareddy** (district median ₹1,300/sq.yd, peaking at ₹19,000/sq.yd along the northwestern ORR edge), alongside the eastern highway corridor in **Yadadri Bhuvanagiri** (median ₹2,000/sq.yd, peak ₹7,400/sq.yd)[6]. At the mandal and village level, the Western IT Corridor and fast-growing suburban rings command the highest official floors outside central Hyderabad: **Kokapet** (Gandipet mandal) leads at ₹47,600/sq.yd, followed by **Manikonda** at ₹40,300/sq.yd, **Puppalguda** at ₹36,800/sq.yd, **Bahadurguda** (Shamshabad mandal) at ₹31,500/sq.yd, **Chandanagar** (Moinabad) at ₹28,400/sq.yd, and **Nizampet** (Bachupally mandal) at ₹26,300/sq.yd[6]. At the aggregate mandal level, **Bachupally** (₹22,600/sq.yd), **Hayathnagar** (₹21,400/sq.yd), and **Gandipet** (₹18,400/sq.yd) form the primary suburban growth arc[6]. Curating 30 to 50 localities across GHMC and these specific peri-urban mandals therefore captures the overwhelming majority of active residential investment variance in Telangana while maintaining high data hygiene[6].

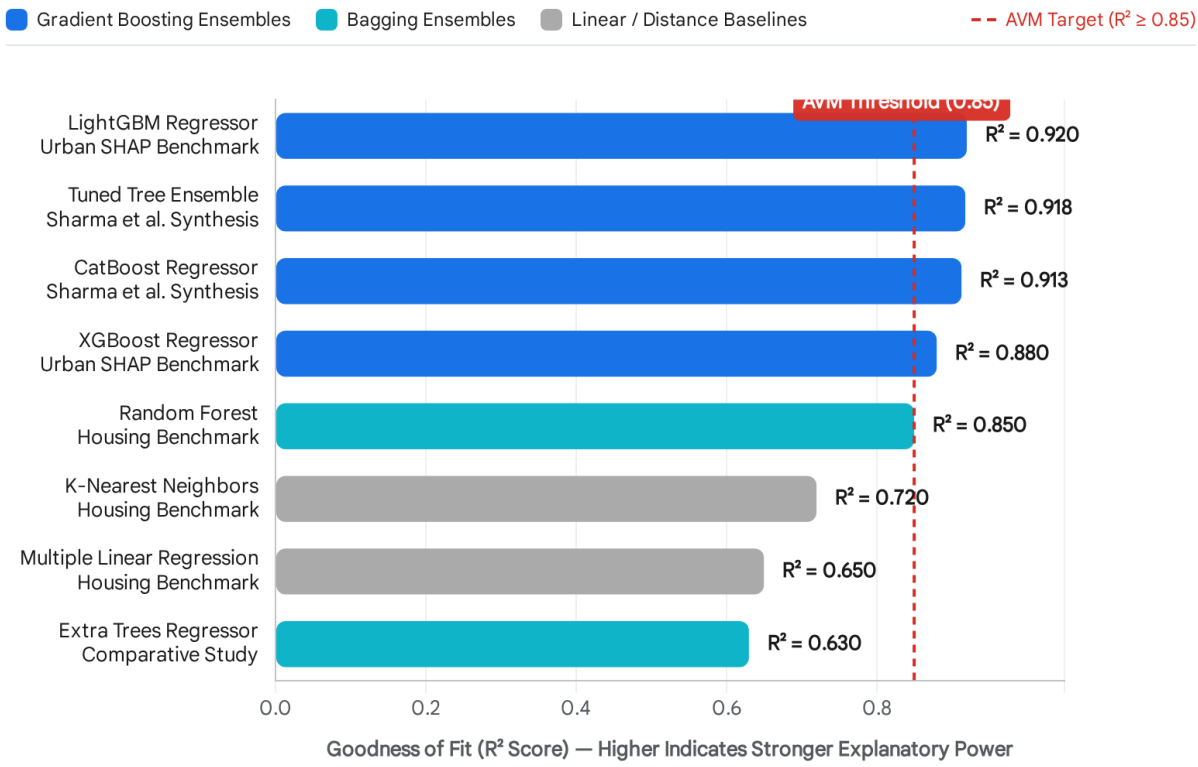
Machine Learning Valuation and Explainable AI (XAI) Scoring Mechanics

Justification of Gradient-Boosted Decision Trees

Conventional property appraisal relies on Multiple Linear Regression (MLR) or static comparative adjustment tables[4]. While linear hedonic models offer transparent coefficient interpretation, they fail to capture complex non-linear interactions and spatial thresholds inherent in urban housing markets[5]. Across post-2021 real estate valuation studies, tree-based ensemble algorithms—specifically XGBoost, LightGBM, CatBoost, and Random Forest—consistently outperform linear regressions, penalized regressions (Ridge, LASSO, Elastic Net), K-Nearest Neighbors (KNN), and Support Vector Regression (SVR)[13]. Deep learning architectures only become competitive when trained on massive multi-modal datasets incorporating street-view imagery or high-frequency financial time series[18]; on structured tabular property datasets, gradient-boosted trees remain the gold standard in both accuracy and computational efficiency[13].

Gradient-Boosted Trees Consistently Outperform Linear and Distance Baselines in **Valuation Accuracy**

Algorithmic Explanatory Power Benchmark (Empirical R^2 Comparison)



Empirical benchmarks across recent real estate valuation literature confirm that tree-boosting ensembles (**LightGBM, XGBoost, and CatBoost**) explain 88% to 92% of price variance, substantially outperforming Random Forest (85%), KNN (72%), and Linear Regression (65%).

Data sources: Nanjing Valuation & SHAP Study (IJOER), Sharma & Ogunleye (MDPI Analytics), Kumar et al. (NIST University), Dabreo et al. (IJRASET)

Mathematically, XGBoost constructs an additive ensemble of K regression trees, where each successive tree f_k corrects the residual errors of the preceding ensemble[26]. The final predicted price per square foot $\hat{y}_i$ for observation i with feature vector x_i is scaled by a shrinkage (learning rate) parameter η[26]:

$$\hat{y}_i = \sum_{k=1}^{K} \eta f_k(x_i), \quad f_k \in \mathcal{F}$$

At boosting iteration t, XGBoost minimizes a regularized objective function that penalizes both the number of terminal leaves T and the squared L_2 norm of leaf weights w_i, preventing overfitting on localized property samples[26]:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} \left(y_i - \left(\hat{y}_i^{(t-1)} + f_t(x_i) \right) \right)^2 + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2$$

During tree construction, XGBoost utilizes the **Weighted Quantile Sketch** algorithm to identify optimal split thresholds across continuous spatial variables and natively learns default branch directions when listing attributes are missing[26]. This missingness-aware architecture aligns directly with the RE-VAL valuation workflow developed by Saiu and Mocci (2024, 2026), which demonstrates that preserving unspecified listing categories rather than discarding incomplete records prevents data loss in heterogeneous real estate portals[4].

Isolating Location Quality and Generating "Why This Score" Bars via TreeSHAP

While XGBoost and LightGBM achieve strong predictive performance ($R^2 = 0.88–0.92$ in recent urban SHAP benchmarks[13]), raw tree ensembles operate as "black boxes" that create a trust deficit among investors and appraisers[1]. To power the **"Why this score"** view on Screen 3, the platform integrates **SHAP (Shapley Additive Explanations)**[4]. Grounded in cooperative game theory, SHAP assigns each feature $j \in \{1, \ldots, M\}$ a unique attribution value ϕ_{ij} representing its exact marginal contribution to shifting the prediction $\hat{y}_i$ away from the metropolitan baseline expectation $\phi_0 = E[f(X)]$ [7]:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^{M} \phi_{ij}$$

Drawing on the **SHAP Location Score** methodology established by Stang et al. (2023, 2024)[7] and the **RE-VAL Bonus/Malus adjustment table** formulated by Saiu and Mocci (2024, 2026)[4], the feature set is partitioned into physical building controls S and locality-level PRICE + Risk variables L [5]. Because Shapley values are strictly additive, the net spatial premium or discount of locality l —the *SHAP Location Score* $\Phi_{\text{Loc.}l}$ —is isolated by summing the Shapley contributions across all locational features $j \in L$ [7]:

$$\Phi_{\text{Loc},l} = \sum_{j \in L} \phi_{lj}$$

When rendering Screen 3, each feature's local SHAP attribution and weighted PRICE contribution is scaled into signed score-point adjustments Δs_{li} around the metropolitan baseline. Positive contributions render as green bonus bars (such as $+2.1$ for proximity to an upcoming metro station or $+1.6$ for high IT-corridor rental yield), whereas negative contributions render as red malus bars (such as -1.3 for an already elevated entry price per square foot or -0.9 for high unsold inventory risk)[4]. These ordered pairs $\{(\text{Feature}_i, \Delta s_{li})\}$ are then passed into a constrained LLM prompt template that generates the single plain-English sentence beneath the chart, guaranteeing that the text explanation remains strictly faithful to the underlying game-theoretic math[4].

Financial Engineering: The NRI Net Return Calculator

Non-Resident Indians (NRIs) represent a vital source of capital in Indian urban real estate, treating property acquisition both as an emotional anchor to the homeland ("home as a project") and a high-conviction financial investment[3]. In southern IT metropolises, real estate developers actively target the diaspora by constructing dual-profile residential enclaves that combine sprawling luxury apartments for returning families with compact one- and two-bedroom units engineered specifically as rental investments catering to white-collar technology employees[23]. However, empirical studies on diaspora finance demonstrate that NRI investors confront substantial hurdles under the Foreign Exchange Management Act (FEMA), including taxation complexities, information asymmetry, unsold inventory delays, and foreign exchange depreciation that can significantly erode dollar- or dirham-denominated returns upon repatriation[3].

Screen 4 (**NRI Net Return Calculator**) addresses this gap by transforming the platform from a domestic price estimator into a cross-border net return simulator[9]. To ensure regulatory accuracy during an academic defense, the calculator incorporates the exact statutory registration structure of Telangana alongside the post-Finance Act (No. 2) 2024 Indian capital gains regime in force in 2026[10].

Statutory Parameters and Mathematical Model (2026 Tax & Regulatory Regime)

When an NRI allocates foreign capital B_{FX} at an initial exchange rate E_0 (yielding a gross INR budget $B_0 = B_{\text{FX}} \times E_0$) over a holding period of T years, the net repatriable return is governed by Telangana acquisition duties, compounded net rental income, post-Budget 2024

capital gains taxation, and annual currency depreciation[9].

Statutory Cost / Tax Parameter	Urban / GHMC Rate (2026)	Rural / Gram Panchayat Rate (2026)	Statutory Authority & NRI Applicability Mechanics
Stamp Duty (Sale Deed)	5.5%[11]	5.5%[25]	Levied on the higher of deed consideration or official government market value under Section 47-A (revised upward from 4.0% in July 2021)[11].
Transfer Duty	1.5%[11]	0.0%[25]	Collected for the Urban Local Body (ULB) or municipal corporation; exempt in rural Gram Panchayats[25].
Registration Fee	0.5%[11]	2.0%[25]	Recording fee at the Sub-Registrar office; shifts from 0.5% in municipal areas to 2.0% in Gram Panchayats[25].
Mutation Fee	0.1%[11]	0.0% (Separate)[11]	Included in the February 2026 Telangana Registration & Stamps Department urban deck, bringing total urban statutory charges to 7.6% (

			7.5% base deed + 0.1% mutation)[11].
Long-Term Capital Gains (LTCG) Tax ($T \geq 2$ Years)	12.5% (No Indexation)[10]	12.5% (No Indexation)[10]	Effective July 23, 2024 (Budget 2024), LTCG was cut from 20% (with indexation) to a flat 12.5% without indexation[10]. **NRIs are excluded from the 20% indexation grandfathering rule**[32].
TDS & Repatriation Rules	Full Sale Value Base (unless Form 13 filed)[31]	Full Sale Value Base (unless Form 13 filed)[31]	Buyers deduct TDS on full sale value unless the NRI obtains a lower-TDS certificate via **Form 13** or claims Sec 54/54F/54EC exemptions[31]; FEMA permits repatriating proceeds of up to 2 residential properties/year[9].

Accounting for Telangana's acquisition charge c_{acq} (7.6% in GHMC/municipal areas or 7.5% in Gram Panchayats)[11], the net initial property asset value acquired from a total outlay B_0 is $P_0 = \frac{B_0}{1+c}$. Over each holding year $t \in \{1, \ldots, T\}$, the property generates gross rental yield y_r escalating at rate g_r. After deducting property management and maintenance expenses m (which are particularly relevant for absentee NRI landlords relying on management agencies[23]) and applying the 30% statutory standard deduction under Section 24(a) at the NRI's effective rental income tax rate τ_r, the net post-tax rental cash flow

in year t is:

$$R_t^{\text{net}} = P_0 \cdot y_r (1+g_r)^{t-1} (1-m)\,(1-0.70\tau_r)$$

At the end of year T, given the locality's projected annual price appreciation rate g_v and exit brokerage cost b_{sell}, the net terminal sale realization before tax is $P_T = P_0(1+g_v)^T(1-b_{\text{sell}})$. For holding periods of two years or longer ($T \geq 2$), the post-Budget 2024 flat 12.5% unindexed Long-Term Capital Gains tax (augmented by applicable surcharge and 4% health and education cess) is deducted from the nominal capital gain[10]:

$$\text{Tax}_{\text{LTCG}} = \max\left(0,\ (P_T - B_0) \times 0.125 \times (1+\text{cess})\right)$$

Finally, because an NRI evaluates wealth in foreign currency, a rising gap between the foreign currency and the Indian Rupee reduces repatriated returns[9]. Modeling an annualized INR depreciation rate δ_{FX} such that the exchange rate in year t is $E_t = E_0(1+\delta_{\text{FX}})^t$, the total terminal wealth in foreign currency $\text{Net Value}_{\text{FX}}(T)$ and the resulting **NRI Net Annualized Return** are computed as:

$$\text{Net Value}_{\text{FX}}(T) = \frac{P_T - \text{Tax}_{\text{LTCG}}}{E_0(1+\delta_{\text{FX}})^T} + \sum_{t=1}^{T} \frac{R_t^{\text{net}}}{E_0(1+\delta_{\text{FX}})^t}$$

$$\text{NRI Net Annualized Return} = \left(\frac{\text{Net Value}_{\text{FX}}(T)}{B_{\text{FX}}}\right)^{\frac{1}{T}} - 1$$

Literature Review Synthesis, Strategic Paper Upgrades, and Comparison Matrix

An academic review panel evaluates a literature survey on whether each architectural module in the system is justified by peer-reviewed evidence and whether the research gap is sharply articulated. Following the five-job framework, the supporting literature must validate: (1) the Indian house price prediction problem, (2) the selection of tree-boosting algorithms (XGBoost/LightGBM), (3) the SHAP location quality score, (4) the explainable AI valuation workflow, and (5) the NRI diaspora real estate investment dimension.

Audit of the Initial Five Papers and Strategic Upgrades

A rigorous bibliographic audit of the initial paper set reveals two strategic opportunities to strengthen the academic defense. First, while **G. Kiran Kumar, D. Malathi Rani, Neeraja Koppula, and Syed Ashraf (2021)**, *"Prediction of House Price Using Machine Learning Algorithms,"* is published in a reputable **IEEE Xplore** conference venue (*2021 5th International Conference on Trends in Electronics and Informatics - ICOEI*, Tirunelveli, India, DOI: 10.1109/ICOEI51242.2021.9452820)[19], the companion Indian paper by **Smith Dabreo, Shaleel Rodrigues, Valiant Rodrigues, and Parshvi Shah (2021)**, *"Real Estate Price Prediction,"* appeared in the *International Journal of Engineering Research & Technology (IJERT)*, a lower-tier student publication[38]. To insulate the literature review from venue-quality criticism, Dabreo et al. (2021) should be replaced or paired with **C. Madhuri, G. Anuradha, and M. Pujitha (2019)**, *"House Price Prediction Using Regression Techniques: A Comparative Study,"* published in **IEEE** (*2019 International Conference on Smart Structures and Systems - ICSSS*, DOI: 10.1109/ICSSS.2019.8882834, 106+ citations)[41], or **J. Manasa, R. Gupta, and N. S. Narahari (2020)**, *"Machine Learning based Predicting House Prices using Regression Techniques,"* published in **IEEE Xplore** (*ICIMIA 2020*, Bengaluru)[2]. Crucially, **Sharma et al. (2024)** explicitly cite and critique **Madhuri et al.** in their own introduction[43], creating a direct bibliographic bridge between Job 1 and Job 2.

Second, to fill the vacant fifth slot justifying the **NRI Net Return Calculator**, the bibliography integrates **S. Irudaya Rajan and Anand P. Cherian (2023)**, *"Remittances and transnational housing among the Indian diaspora: Home as a project,"* published in the *Handbook on Home and Migration* (Edward Elgar Publishing, pp. 647–659)[3], alongside **Aurélie Varrel's** foundational urban study *"NRIs in the City: Identifying International Migrants' Investments in the Indian Urban Fabric"*[23] and recent empirical surveys on NRI real estate return perceptions in South India and FEMA regulatory friction across the GCC–India corridor[3].

Deep-Dive Executive Summaries of the Core Papers

To ensure complete readiness for technical questioning regarding each paper's dataset, methodology, quantitative results, and limitations, the essential mechanics of the literature set are synthesized below in narrative form.

For **Job 1 (Proving the Problem Exists in India)**, **Kumar, Rani, Koppula, and Ashraf (2021)** (*IEEE ICOEI*) formulate Indian residential property valuation as a supervised machine learning pipeline implemented in Python (Pandas, NumPy, Scikit-learn)[12]. Evaluating Random Forest against linear and multiple regression models across physical and locational property attributes, the authors demonstrate that ensemble decision trees capture complex non-linear interactions far more effectively than linear baselines, achieving lower RMSE and higher R^2[8]. Complementing this work, the recommended IEEE upgrade by **Madhuri, Anuradha, and Pujitha (2019)** (*IEEE ICSSS*) compares Multiple Linear Regression, Ridge, LASSO, Elastic Net, and Gradient Boosting (AdaBoost/XGBoost) to assist buyers and sellers in estimating property prices without broker bias[41]. While Madhuri et al. confirm that boosting ensembles outperform

penalized linear regressions, Sharma et al. (2024) point out that Madhuri et al. relied on default algorithm parameters without hyperparameter tuning[14]. If an examiner specifically asks about **Dabreo, Rodrigues, Rodrigues, and Shah (2021)** (*IJERT*), that study similarly compared regression and tree algorithms on Indian metropolitan housing data to build a web-based price estimator[38]. Across all three Indian studies, the overarching limitation is identical: they output isolated, static price predictions in a tabular interface without geospatial polygon mapping, SHAP explainability, forward-looking infrastructure scoring, or NRI tax adjustments.

For **Job 2 (Justifying the XGBoost Model)**, **Hemlata Sharma, Hitesh Harsora, and Bayode Ogunleye (2024)** published *"An Optimal House Price Prediction Algorithm: XGBoost"* in **MDPI** *Analytics* (Vol. 3, No. 1, pp. 30–45; also archived on **arXiv**: 2402.04082, 16 pages)[14]. Recognizing that property value is heavily shaped by surrounding neighborhood characteristics alongside physical attributes, the authors conduct a systematic experimental benchmark of hyperparameter-tuned XGBoost against Multiple Linear Regression, Ridge, LASSO, Elastic Net, Random Forest, and AdaBoost on the Ames, Iowa housing dataset[14]. Their results establish that tuned XGBoost is the superior model across accuracy (R^2), error minimization ($RMSE$, MAE), computational scalability, and feature-importance clarity[14]. However, Sharma et al. evaluate a single US city dataset and focus strictly on predicting raw transaction prices rather than constructing an interactive, map-based multi-factor investment score for an Indian metropolis[14].

For **Job 3 (Justifying the Location Score via Explainable AI)**, **Moritz Stang, Bastian Krämer, Marcelo Cajias, and Wolfgang Schäfers (2023 / 2024)** authored *"Changing the Location Game – Improving Location Analytics with the Help of Explainable AI"* and *"Isolating Location Value Using SHAP and Interaction Constraints"* (*Journal of Real Estate Research*, 2024, Vol. 46, No. 4, pp. 421–443)[7]. Addressing the inability of traditional automated valuation models (AVMs) to cleanly separate micro-location quality from structural building traits, Stang et al. train an XGBoost regressor on residential property data in the London, UK region and apply SHAP with tree interaction constraints[7]. By summing the Shapley values of all spatial variables, they extract the **SHAP Location Score**, proving that game-theoretic additive decomposition yields a rigorous, spatially continuous metric of location value aligned with hedonic pricing theory[7]. The limitation of Stang et al. is that their score measures static hedonic location value in a mature European market (London) without accounting for emerging-market infrastructure catalysts, rental yield differentials, environmental/legal risk penalties, or diaspora repatriation mechanics[7].

For **Job 4 (Justifying the "Why This Score" Explanation Workflow)**, **Valeria Saiu and Matteo Mocci (2024 / 2026)** (University of Cagliari) developed the **RE-VAL (REal-estate VALuation)** framework in *"Explainable AI for Urban Real-Estate Prediction: A Machine-Learning Framework for Urban Decision Support"* (**MDPI** *Sustainability* 2024, 16(12), 5120 / *Urban Science* 2026, 10(6), 315)[4]. Evaluating fragmented Italian urban property markets where digital portal listings suffer from uneven and missing technical specifications, Saiu and Mocci combine missingness-aware preprocessing with gradient-boosted trees and SHAP[4]. Their central contribution is translating abstract SHAP outputs into auditable monetary **Bonus/Malus adjustment tables** aligned with professional comparative appraisal methodology (D'Amato &

Kauko, 2017)[4]. This allows investors and appraisers to inspect context-sensitive positive and negative value drivers for any property[4]. While RE-VAL directly validates Screen 3's positive/negative adjustment bars and missing-data resilience, it stops at appraisal auditing in Italy and does not provide an interactive green-to-red investment choropleth map, an LLM plain-English synthesis layer, or an NRI return calculator[4].

For **Job 5 (Justifying the NRI Investment Angle)**, **S. Irudaya Rajan and Anand P. Cherian (2023)** examine *"Remittances and transnational housing among the Indian diaspora: Home as a project"* (*Handbook on Home and Migration*, Edward Elgar, pp. 647–659)[3], complemented by **Aurélie Varrel's** empirical investigation *"NRIs in the City: Identifying International Migrants' Investments in the Indian Urban Fabric"*[23]. Rajan and Cherian demonstrate that India is the world's largest remittance recipient ($84B+ annually[51]) and that residential real estate serves as the primary capital sink for overseas Indians, blending emotional attachment with long-term wealth creation[3]. Varrel further documents how Indian developers actively market metropolitan IT-corridor apartments abroad as high-yield rental and speculative assets targeting absentee NRI landlords[23]. Crucially, both Rajan & Cherian (2023) and recent diaspora finance studies highlight that NRI property investors face severe information deficits, exchange-rate volatility, and regulatory/FEMA compliance barriers, underscoring an urgent need for transparent financial decision-support tools[3]. Because these sociological and economic studies offer no software implementation or quantitative valuation engine, Screen 4 directly operationalizes their findings.

The One-Line Literature Comparison Matrix

Paper & Venue	Method	Result	What the Project Takes	Gap the Project Fills
Kumar et al. (2021)[19] *IEEE ICOEI* (Job 1A)	Random Forest vs. Linear Regression on Indian housing data[19].	Random Forest captures non-linear interactions best[8].	Validates ML price prediction on Indian residential attributes[19].	Static tabular price output; no GIS map, no XAI explanations, and no investment score.
Madhuri et al. (2019)[41] *IEEE ICSSS* (Job 1B — *IEEE*	Compares MLR, Ridge, LASSO, Elastic Net, and Boosting on	Ensemble boosting beats penalized linear models[41].	Proves standard linear models underfit Indian property	Used default hyperparameters[14]; predicts asking price only without

Swap) *(Replaces* ***Dabreo et al. 2021****, IJERT)*[38]	Indian data[41].		dynamics[41].	spatial scoring or NRI view.
Sharma et al. (2024)[14] *MDPI Analytics / arXiv* (Job 2)	Tuned XGBoost vs. 6 regression & ensemble baselines[14].	Tuned XGBoost achieves highest accuracy (R^2) and scalability[14].	Adopts hyperparameter-tuned XGBoost / LightGBM as the core valuation engine[13].	Non-Indian dataset (Ames, Iowa)[26]; predicts raw house price rather than a map-based investment score.
Stang et al. (2023/2024)[7] *J. Real Estate Research* (Job 3)	XGBoost + TreeSHAP with interaction constraints on London data[7].	Extracts an additive **SHAP Location Score** isolating spatial value[7].	Uses additive SHAP decomposition to quantify locality-level spatial scores[7].	Evaluates static London location value only[7]; lacks upcoming infrastructure, rental yield, and risk penalties.
Saiu & Mocci (2024/2026)[4] *MDPI Sustainability / Urban Sci.* (Job 4)	**RE-VAL** workflow: missingness-aware ML + SHAP Bonus/Malus tables[4].	Converts black-box SHAP outputs into auditable monetary $+/-$ adjustments[4].	Implements the **"Why This Score"** $+/-$ bar chart and preserves incomplete portal listings[4].	Italian appraisal focus[4]; no green-to-red investment choropleth UI or plain-English LLM synthesis.
Rajan & Cherian (2023)[3] *Edward Elgar* & **Varrel (2016)**[23]	Empirical analysis of NRI remittances, FEMA friction, and urban IT housing	Real estate is the top NRI asset class but suffers from severe information &	Justifies the **NRI Net Return Calculator** accounting for FX, FEMA, and	Purely qualitative/economic analysis; provides no predictive software, map

(Job 5 — *NRI Addition*)	investment[3].	FX asymmetry[3].	Indian taxes[3].	interface, or tax/FX calculator.

Execution Roadmap and Methodological Safeguards

Ranked Six-Stage Engineering Build Order

To deliver a resilient Minimum Viable Product (MVP) without scope overextension, engineering execution follows a strict six-stage dependency sequence. The initial four stages establish the core spatial dataset for roughly 40 localities, the gradient-boosted price model, the green-to-red investment map, and the SHAP explanation panel; only after this foundation is verified are the NRI financial simulator and multi-locality comparison screen layered onto the application.

Build Rank	Engineering Milestone	Target Technical Scope & Deliverables	Core Validation Metric / Dependency
Stage 1	**Curated Dataset for ~40 Localities**	Build a clean PostgreSQL/PostGIS dataset covering 30 to 50 strategic localities across GHMC, Rangareddy, Medchal-Malkajgiri, and Sangareddy (e.g., Kokapet, Manikonda, Puppalguda, Gachibowli, Madhapur, Kondapur, Bachupally, Nizampet, Shamshabad, Chandanagar, Hayathnagar, Uppal, Kompally)[6]; attach OpenStreetMap	Zero coordinate nulls; verified spatial joins (ST_Distance) to Metro, ORR, RGIA airport, and IT job hubs.

		distances and official Telangana Registration unit rates[6].	
Stage 2	**Tabular Price Prediction Model**	Train and cross-validate XGBoost and LightGBM regressors against Multiple Linear Regression and Random Forest baselines on the Stage 1 feature store[13].	Held-out test evaluation logging exact MAE, $RMSE$, and R^2 [12].
Stage 3	**Composite Score + Choropleth Map**	Compute normalized PRICE + Risk sub-scores, combine with model valuation residuals to scale the $1–10$ investment score, and render the interactive green-to-red React + MapLibre GL polygon map with **Budget**, **Goal**, and **Property Type** filter chips (Screens 1 & 2).	Sub-second vector polygon re-coloring upon filter chip selection.
Stage 4	**"Why This Score" XAI Panel**	Wire shap.TreeExplainer outputs into signed Bonus/Malus horizontal bars[4] and connect the	Exact additivity check ($\sum \phi_{ij} + \phi_0 =$) with zero LLM numeric hallucination[4].

		deterministic/LLM plain-English sentence synthesizer (Screen 3).	
Stage 5	**NRI Net Return Calculator**	Implement the parametric cross-border return engine incorporating Telangana's 7.6% urban registration duty, the post-Budget 2024 12.5% unindexed LTCG tax, rental taxation, and FX depreciation (Screen 4)[9].	Verified mathematical parity against manual post-Budget 2024 tax schedules[11].
Stage 6	**Multi-Locality Compare Screen**	Construct the side-by-side tabular comparison view for 2 to 3 user-selected localities across all PRICE metrics, risk deductions, and NRI net yields (Screen 5).	Synchronized state management between map selection and comparison matrix.

Proactive Methodological Disclosures

In an academic defense, pre-emptively stating the empirical boundaries of the data pipeline demonstrates architectural maturity and neutralizes common panel criticisms before they are raised. First, because real estate transactions in India are recorded via physical deed registrations rather than a centralized, tick-by-tick electronic exchange, genuine "real-time" pricing does not exist publicly; accordingly, the system describes its data cadence as **regularly refreshed (on a weekly or monthly batch cycle)** rather than claiming live streaming updates. Second, commercial property portals publish seller *asking prices*, which systematically exceed closed transaction values due to negotiation cushions[16]. Stating this limitation explicitly—and

mitigating it by pairing retail asking prices with official Telangana Registration and Stamps Department unit rates (registration.telangana.gov.in) as a statutory valuation floor[6]—transforms a potential data weakness into a methodological strength.
Third, because aggressive automated web scraping can violate commercial portal terms of use and result in brittle pipelines, the MVP prioritizes open geospatial repositories (OpenStreetMap), public metro and highway alignments, official Telangana government guidance value schedules, and a curated, audited benchmark dataset for its 40 target localities[6]. Finally, the presentation must never assert unmeasured accuracy percentages; predictive performance is reported strictly through empirical out-of-sample regression metrics (MAE, $RMSE$, and R^2) measured after cross-validation testing[12].

Potential Panel Critique	**Underlying Market / Technical Reality**	**Proactive Defense & Architectural Mitigation**
"Real-Time" Pricing Claims	Public tick-by-tick live property transaction feeds do not exist in Indian real estate.	Describe the pipeline as **"regularly refreshed" (weekly or monthly batch updates)** to reflect realistic market velocity.
Asking Prices vs. Sold Prices	Commercial listing portals display seller asking prices that include a negotiation buffer[16].	Explicitly state this limitation and anchor every locality with **Telangana Registration & Stamps guidance unit rates** as a second statutory source[6].
Portal Scraping & Terms of Use	Automated scraping of commercial real estate sites can violate terms of service.	Prioritize open geospatial data (OpenStreetMap), government guidance schedules (registration.telangana.gov.in), and a clean, curated 40-locality dataset[6].
Unverified Accuracy Claims	Quoting arbitrary percentage accuracy (e.g., "95% accurate") is scientifically invalid for continuous regression.	Report only empirically measured MAE, $RMSE$, **and** R^2 values obtained on held-out test folds after model

		evaluation[12].

Comprehensive Panel Defense Q&A Compendium

The following question-and-answer matrices prepare the defense presentation to address rigorous technical, econometric, spatial, and financial questions posed by an academic review committee.

Category 1: Machine Learning, XAI & Scoring Mechanics

Likely Panel Question	Defensible, High-Density Technical Answer
Q1. Why does the system use XGBoost or LightGBM instead of Deep Learning (such as Neural Networks or LSTMs)?	Residential property valuation across 40 to 50 localities with structured spatial-economic features is a classic tabular regression problem[14]. Systematic literature reviews (including Sharma et al., 2024 and post-2023 PRISMA syntheses) demonstrate that gradient-boosted decision trees consistently outperform deep neural networks on structured tabular datasets of moderate size[14]. Deep learning requires tens of thousands of multi-modal samples (such as street-view imagery) to avoid overfitting[18], whereas XGBoost incorporates L_1 / L_2 regularization and exact TreeSHAP explainability with rapid training convergence[7].
Q2. How does the architecture prevent SHAP explanations and the LLM from hallucinating financial advice?	The LLM has zero involvement in computing prices, scores, or feature attributions. TreeSHAP deterministically calculates exact game-theoretic Shapley values (ϕ_{ij}) directly from the trained XGBoost decision trees[4]. Following Saiu and Mocci's (2024/2026) RE-VAL Bonus/Malus methodology, these exact numeric values populate the diverging bar chart[4]. The LLM is strictly constrained via a template to

	convert the top pre-computed (feature, signed_score_delta) tuples into a single grammatically fluent summary sentence.
Q3. A high predicted price per sq.ft does not mean a locality is a good investment. How does the model distinguish an expensive area from a high-return area?	That distinction is the central innovation of the **PRICE + Risk** scoring layer. A mature luxury locality in central Hyderabad commands a high baseline price per sq.ft, which SHAP flags as a negative entry-price adjustment (e.g., -1.3 *high price already*)[4]. The 1–10 Investment Score combines the model's valuation residual (whether current prices are undervalued relative to fundamentals) with forward-looking **Price Growth**, **Rental Yield**, and upcoming **Infrastructure** proximity[5]. Consequently, an emerging hub near a planned metro corridor or ORR interchange scores higher on investment potential than an already saturated luxury zone.
Q4. How does the pipeline handle incomplete or missing attributes in property records?	In alignment with the RE-VAL framework by Saiu and Mocci (2024/2026), digital property records frequently exhibit uneven reporting of secondary specifications[4]. Rather than discarding observations via listwise deletion or applying naive mean imputation, XGBoost natively handles missing values via sparsity-aware split finding (learning the optimal default branch direction for missing attributes)[4], while categorical missingness is preserved as an explicit informative signal[4].

Category 2: Data Engineering, Spatial Scope & Hyderabad Specifics

Likely Panel Question	**Defensible, High-Density Technical Answer**

Q5. Property portals show inflated asking prices rather than registered sale prices. How does the dataset account for this discrepancy?	This is a structural limitation of Indian real estate portals, which the architecture mitigates through **dual-source price anchoring**. Alongside retail asking prices, the database integrates official **Telangana Registration and Stamps Department unit rates** (registration.telangana.gov.in)[6]. Under Section 47-A of the Indian Stamp Act, these government guidance values represent the statutory minimum floor for stamp duty assessment[6]. Modeling both the retail asking rate and the statutory guidance floor captures the true valuation corridor of each locality[6].
Q6. Why limit the Minimum Viable Product to 30–50 localities instead of modeling every village in Telangana?	Empirical analysis of the 2026 Telangana guidance value dataset across 9,851 villages demonstrates that 85.1% of villages statewide have flat, rural land rates below ₹2,000/sq.yd[6]. Virtually the entire residential investment gradient is concentrated in GHMC (median ₹33,500/sq.yd) and the urbanizing mandals of Rangareddy, Medchal-Malkajgiri, and Sangareddy (such as Kokapet at ₹47,600/sq.yd and Manikonda at ₹40,300/sq.yd)[6]. Curating 40 core localities across these corridors captures the overwhelming majority of active investment volume while ensuring high data quality and zero scraping violations.
Q7. Why select PostgreSQL/PostGIS and MapLibre over a standard document database and Google Maps API stack?	First, PostGIS provides native spatial indexing (R-tree over GiST) and geometric functions (ST_Distance, ST_Intersects) required to compute exact distances from locality polygons to OpenStreetMap metro lines, ORR interchanges, and IT employment hubs. Second, MapLibre GL is an open-source, WebGL-accelerated vector rendering engine that avoids commercial per-tile paywalls while delivering smooth 60-fps choropleth polygon shading in

	React.

Category 3: NRI Financial Calculator & Literature Defense

Likely Panel Question	Defensible, High-Density Technical Answer
Q8. What specific Indian tax and Telangana registration rules are embedded in the NRI Net Return Calculator?	The calculator implements the exact statutory regime in force in 2026: (1) On acquisition in urban/GHMC areas, it deducts Telangana's **7.6% registration and stamp duty charge** (5.5% Stamp Duty + 1.5% Transfer Duty + 0.5% Registration Fee + 0.1% Mutation Fee) calculated on the higher of consideration or government market value[11]; (2) On exit after 24 months, it applies the **post-Budget 2024 Long-Term Capital Gains (LTCG) rate of 12.5% without indexation** (noting that NRIs are legally excluded from the 20% indexation grandfathering option)[10]; and (3) It converts net INR cash flows into the NRI's base currency factoring in annualized exchange-rate depreciation and FEMA repatriation rules[3].
Q9. What is the single overarching research gap that this project fills compared to the five papers in the literature review?	Existing Indian studies (Kumar et al., 2021; Madhuri et al., 2019; Dabreo et al., 2021) predict static property prices in tabular isolation without spatial visualization or explainability[19]. Conversely, international XAI valuation studies (Sharma et al., 2024; Stang et al., 2024; Saiu & Mocci, 2026) demonstrate XGBoost and SHAP location scoring on Western/European markets (Iowa, London, Italy)[4] without forward-looking investment scoring or diaspora tax mechanics. This platform bridges both gaps: it is the first framework

	to combine XGBoost + TreeSHAP Bonus/Malus explainability with an interactive PostGIS/MapLibre investment choropleth map tailored to Hyderabad/Telangana and integrated with a statutory NRI net-return simulator.

Conclusions and Strategic Recommendations

Academic review panels evaluate numerous generic house-price regression notebooks every academic cycle, most of which terminate at standard root-mean-square error plots on static datasets[8]. The primary differentiator of the proposed platform lies in shifting the analytical paradigm from isolated point-price prediction to explainable geospatial investment scoring[4]. By synthesizing historical price momentum, rental yield, upcoming transit infrastructure, transport connectivity, employment hub gravity, and environmental or inventory risk penalties into a unified 1-to-10 index (PRICE + Risk), the system transforms complex econometric outputs into an immediately legible green-to-red choropleth map tailored to the urban morphology of Hyderabad and Telangana[5].

To maximize the rigor of the literature review defense, the bibliographic upgrade should be finalized immediately prior to presentation. Retaining **Kumar et al. (2021)** (*IEEE ICOEI*)[19] while substituting or pairing the lower-tier *IJERT* paper (**Dabreo et al., 2021**)[39] with the highly cited **IEEE** comparative study by **Madhuri et al. (2019)** (*IEEE ICSSS*)[41] ensures that the Indian valuation baseline meets strict peer-review standards and links directly to the literature synthesis in **Sharma et al. (2024)** (*MDPI Analytics / arXiv*)[14]. Pairing **Stang et al. (2023/2024)** (*Journal of Real Estate Research*)[7] and **Saiu & Mocci (2024/2026)** (*MDPI Sustainability / Urban Science*)[4] with the diaspora housing scholarship of **Rajan & Cherian (2023)** (*Edward Elgar*)[3] and **Varrel (2016)**[23] establishes a complete, five-pillar theoretical foundation in which every screen of the software corresponds directly to published academic literature.

Finally, grounding the 40-locality Minimum Viable Product in official **Telangana Registration and Stamps Department unit rates** (registration.telangana.gov.in) alongside OpenStreetMap geospatial vectors provides a decisive methodological moat[6]. This dual-source design insulates the project against criticisms regarding portal terms-of-service violations and asking-price inflation, while enabling the **NRI Net Return Calculator** to model authentic Section 47-A stamp duty floors (7.6% urban / 7.5% rural), post-Budget 2024 unindexed Long-Term Capital Gains taxation (12.5%), and foreign exchange repatriation effects with institutional precision[9].

Works cited

1. Artificial Intelligence (AI) in Real Estate Valuation - OMR Publication, https://omrpublication.com/wp-content/uploads/2026/08/OMRP-JMID-2026-004

3.pdf
2. House Pricing Prediction using ML Algorithm - A Comparative Analysis, https://ijsrst.com/IJSRST229242
3. (PDF) Diaspora Finance and Economic Connectivity: A Study of NRI, https://www.researchgate.net/publication/406975404_Diaspora_Finance_and_Economic_Connectivity_A_Study_of_NRI_Investment_from_GCC_Countries_to_India
4. Explainable AI for Urban Real-Estate Prediction: A Machine ... - MDPI, https://www.mdpi.com/2413-8851/10/6/315
5. Beyond Prediction: A Generalized Explainable Hybrid Econometric, https://www.mdpi.com/2813-8090/3/3/15
6. Telangana Open Plot Rates 2026: Guidance vs Market Value - Assetly, https://assetlyhq.com/blog/telangana-open-plot-urban-land-rates-2026/
7. Improving Location Analytics with the Help of Explainable AI, https://www.researchgate.net/publication/374312705_Changing_the_Location_Game_-_Improving_Location_Analytics_with_the_Help_of_Explainable_AI
8. Predictive Housing Price Analysis in India - Machine Learning - Scribd, https://www.scribd.com/document/931226217/Sip-Presentation-Nishan
9. NRI Real Estate Investment In India: Everything You Must Know, https://www.gharpedia.com/blog/nri-real-estate-investment-in-india/
10. Capital Gains Tax on Property in India 2026 - Godrej Castillo, https://www.godrejscastillo.com/investment/capital-gains-tax-property-india.html
11. Telangana 7.6% Duty: Price or Market Value - PropNewz, https://www.propnewz.com/blog/telangana-sale-deed-7-6-percent-higher-of-consideration-or-market-value-section-47-a-deficit-duty-hyderabad-buyer-2026-11-04
12. House Price Prediction Using Machine Learning - ijarsct, https://ijarsct.co.in/Paper29515.pdf
13. House Price Prediction in Nanjing Based on Machine Learning and, https://ijoer.com/article-details/house-price-prediction-in-nanjing-based-on-machine-learning-and-shap-interpretability
14. An Optimal House Price Prediction Algorithm: XGBoost, https://www.researchgate.net/publication/377065514_An_Optimal_House_Price_Prediction_Algorithm_XGBoost
15. Incorporating Spatial Autocorrelation in Machine Learning Models, https://www.researchgate.net/publication/359808809_Incorporating_Spatial_Autocorrelation_in_Machine_Learning_Models_Using_Spatial_Lag_and_Eigenvector_Spatial_Filtering_Features
16. Improving hedonic housing price models by integrating optimal, https://www.researchgate.net/publication/373190649_Improving_hedonic_housing_price_models_by_integrating_optimal_accessibility_indices_into_regression_and_random_forest_analyses
17. Real Estate Price Prediction Using Machine Learning Techniques, https://www.researchgate.net/publication/417057162_Real_Estate_Price_Predictio

n_Using_Machine_Learning_Techniques_A_Review_and_a_Proposed_Explainable_Framework
18. Machine Learning in Real Estate Price Prediction | Systematic Review, https://www.scribd.com/document/974306818/Ephrems-s-Systematic-Review
19. AI-Powered Real Estate Price Predictor - IJSREM, https://ijsrem.com/download/ai-powered-real-estate-price-predictor/fulltext.pdf
20. Vasavi Gannina | Portfolio, https://vasavigannina.com/
21. House Price Prediction Analysis using Machine Learning, https://ijcrt.org/papers/IJCRT2204637.pdf
22. HOUSE PRICE PREDICTION USING MACHINE LEARNING - TIJER, https://tijer.org/tijer/papers/TIJER2304445.pdf
23. NRIs-in-the-City-Identifying-International-Migrants-Investments-in, https://www.researchgate.net/publication/304592848_NRIs_in_the_City_Identifying_International_Migrants'_Investments_in_the_Indian_Urban_Fabric/fulltext/577e561f08aeaa6988b0c1e8/NRIs-in-the-City-Identifying-International-Migrants-Investments-in-the-Indian-Urban-Fabric.pdf
24. Real estate valuation with multi-source image fusion and enhanced, https://pmc.ncbi.nlm.nih.gov/articles/PMC12088074/
25. Plot Registration Cost Calculator Telangana (Stamp Duty) | YIH, https://www.youngindiahousing.com/tools/plot-registration-cost-calculator/
26. Artificial intelligence models in real estate forecasting: a machine, https://www.emerald.com/ecam/article/33/6/4833/1259788/Artificial-intelligence-models-in-real-estate
27. (PDF) LSTM-GNN Synergy: A New Frontier in Stock Price Prediction, https://www.researchgate.net/publication/386407889_LSTM-GNN_Synergy_A_New_Frontier_in_Stock_Price_Prediction
28. Improving Location Analytics with the Help of Explainable AI, https://ideas.repec.org/a/taf/rjerxx/v46y2024i4p421-443.html
29. Matteo Mocci - Google Académico, https://scholar.google.es/citations?user=fNZGTukAAAAJ&hl=pt-PT
30. Remittances and transnational housing among the Indian diaspora, https://www.researchgate.net/publication/371596837_Remittances_and_transnational_housing_among_the_Indian_diaspora_home_as_a_project
31. How US-based NRI can Plan Taxes before selling Property in India, https://investmates.io/blog/nri-indian-property-sale-tax-planning
32. Capital Gains Tax on Property Sale in India 2026 - Godrej Beacon, https://www.godrejsbeacon.com/blog/capital-gains-tax-property.html
33. Stamp Duty and Registration Charges in Telangana 2026 - Willjini, https://www.willjini.com/blog/stamp-duty-and-registration-charges-in-telangana-2026/
34. SELLING PROPERTY IN INDIA? HERE'S HOW NRIs CAN AVOID, https://trilegal.com/news-insights/news-insights-thoughtleadership-himanshu-sinha-selling-property-in-india-heres-how-nris-can-avoid-excess-tds/
35. house price prediction using machine learning algorithm - IJSDR, https://ijsdr.org/papers/IJSDR2404200.pdf

36. 2021 5th International Conference on Trends in Electronics and, https://www.proceedings.com/content/059/059274webtoc.pdf
37. Prediction of House Price Using Machine Learning Algorithms | PDF, https://www.scribd.com/document/998238649/Prediction-of-House-Price-Using-Machine-Learning-Algorithms
38. [PDF] Real Estate Price Prediction - Semantic Scholar, https://www.semanticscholar.org/paper/Real-Estate-Price-Prediction-Dabreo-Rodrigues/5ecd638ffae62e308b8d5843c9140460199b85bd
39. Real Estate Price Prediction - IJERT, https://www.ijert.org/real-estate-price-prediction
40. Real Estate Price Prediction System, https://ijarsct.co.in/A3394.pdf
41. House Price Prediction Using Regression Techniques, https://www.semanticscholar.org/paper/House-Price-Prediction-Using-Regression-Techniques%3A-Madhuri-Anuradha/252045b616ef9311fe7438d6ff02978aa1888f62
42. House Price Prediction | International Journal of Computer Sciences, https://ijcseonline.org/index.php/j/article/view/6122
43. An Optimal House Price Prediction Algorithm: XGBoost, https://encyclopedia.pub/entry/53852
44. An Optimal House Price Prediction Algorithm: XGBoost - MDPI, https://www.mdpi.com/2813-2203/3/1/3
45. Anand Panamthottam Cherian - Google Scholar, https://scholar.google.com/citations?user=ERF_COYAAAAJ&hl=en
46. Handbook on Home and Migration - Edward Elgar Publishing, https://www.e-elgar.com/shop/gbp/handbook-on-home-and-migration-9781049405483.html
47. an analysis on the perceptions of nris on investment in real estate, https://www.researchgate.net/publication/373976268_AN_ANALYSIS_ON_THE_PERCEPTIONS_OF_NRIS_ON_INVESTMENT_IN_REAL_ESTATE
48. View of Property Price Prediction Using Machine Learning, https://jisem-journal.com/index.php/journal/article/view/7980/3645
49. [2402.04082] An Optimal House Price Prediction Algorithm: XGBoost, https://arxiv.org/abs/2402.04082
50. (PDF) Explainable AI for Urban Real-Estate Prediction: A Machine, https://www.researchgate.net/publication/405979623_Explainable_AI_for_Urban_Real-Estate_Prediction_A_Machine-Learning_Framework_for_Urban_Decision_Support
51. 37. Home and homemaking practices among skilled Indian migrants, https://dspace.library.uu.nl/bitstreams/7607f977-7de2-4284-bceb-0a295516dc2e/download
52. House Price Prediction: Comparative Analysis of Regression-Based, https://www.ijraset.com/research-paper/house-price-prediction-comparative-analysis-of-regression-based-machine-learning-algorithms
53. Property Capital Gains Tax 2026: 12.5% Flat vs 20% with Indexation, https://fincalculator.in/blog/property-capital-gains-tax-indexation-vs-12-5-guide

54. [PDF] The Impacts of Open Data and eXplainable AI on Real Estate, https://www.semanticscholar.org/paper/The-Impacts-of-Open-Data-and-eXplainable-AI-on-Real-Neves-Aparicio/aa0357b542acee388c13b76a339ab42936a67025